

Requirements Elicitation is the process of discovering, gathering, and understanding the needs and constraints for a new or modified software system from its various stakeholders. It is a critical initial phase in Software Requirements Engineering.

Here is an explanation broken down into questions and answers, focusing on key resources and techniques.

1. What are the Resources for Requirements Elicitation?

Answer: Elicitation resources are the primary sources of information and knowledge used to gather requirements. They are typically categorized into **Stakeholders** and **Existing Documentation/Systems**.

Resource Category	Description	Small Example
Stakeholders	Individuals or groups who are affected by the system or have a direct or indirect influence on the requirements.	For a New Library System : Librarians (end-users), Library Director (sponsor), Patrons (external users), and IT staff (technical experts).
Existing Documentation	Current business documents, policy manuals, system specifications, reports, and procedures that define the current state or business rules.	The library's "Book Check-Out Policy Manual" details the rules for lending periods, late fees, and patron registration, which directly impacts the new system's requirements.
Existing Systems	The current, possibly non-computerized or older software system, which shows how tasks are currently performed.	Observing the existing manual card catalog and checkout process highlights requirements like <i>maintaining a book's history</i> and <i>patron identification methods</i> .

2. What are the Most Common Elicitation Techniques?

Answer: Elicitation techniques are the systematic methods used by analysts to extract requirements from the resources. The choice of technique depends on the project, stakeholders, and available time.

Technique 1: Interviews

Q: What is the **Interview** technique and when is it most effective?

A: An interview is a structured or unstructured conversation with a stakeholder to understand their needs, tasks, and system expectations.

Effectiveness: Best for in-depth understanding of individual perspectives and complex processes, especially when dealing with a small, critical group of stakeholders.

Small Example:

- **Analyst to Librarian:** "Can you walk me through the steps you currently take when a patron returns a book that is overdue, and what information do you need readily available

on the screen to calculate the fine?"

- *Requirement Discovered:* **The system must automatically calculate late fees based on the return date and patron type.**

Technique 2: Workshops (JAD/Requirements Workshops)

Q: What is a **Requirements Workshop** and what is its main advantage?

A: A requirements workshop (often Facilitated or Joint Application Development - JAD) is a structured, intensive meeting bringing together a diverse group of key stakeholders and analysts to collaboratively define, review, and agree upon requirements.

Main Advantage: It promotes consensus and resolves conflicting requirements quickly by having all critical parties in the same room.

Small Example:

- **Scenario:** A workshop with the Library Director (wants high revenue from fines) and a Librarian (wants patron satisfaction) to define the late fee rules.
- *Conflict Resolution:* **Agreement is reached that fines cap at \$10 to balance revenue and patron retention, and the system must enforce this cap.**

Technique 3: Observation (Contextual Inquiry)

Q: What is the **Observation** technique and what does it reveal that other techniques might miss?

A: Observation involves watching stakeholders perform their tasks in their natural work environment, often including taking notes or video recording.

What it Reveals: It uncovers "tacit knowledge"—the unstated, habitual, or unconscious requirements that stakeholders may forget to mention in an interview because they are so routine.

Small Example:

- **Observation:** The analyst observes a librarian placing returned books on a sorting cart, then picking up a separate list to check for books on hold, and physically moving those to a "holds" shelf.
- *Requirement Discovered:* **The system must display an immediate alert when a returned book is on hold for another patron, prompting the librarian to flag it before it goes to the main sorting cart.**

Technique 4: Prototyping

Q: How is **Prototyping** used as a requirements elicitation technique?

A: Prototyping involves creating a preliminary, working model (a mock-up or wireframe) of the system's user interface or key functionality to obtain early feedback from users.

Usage: It converts abstract ideas into tangible forms, allowing stakeholders to "try before they buy" and refine their needs based on interaction.

Small Example:

- **Action:** The analyst shows the librarian a wireframe of the patron search screen, including fields for name, ID, and recent activity.
- *Feedback: Librarian:* "I need to see the patron's current number of checked-out books and any outstanding fines right on the main search screen—not on a separate page—because that's what I check first."

- *Requirement Refined:* **The Patron Search screen must prominently display the count of currently checked-out items and total outstanding fines.**